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Cement Top Job Utilizing an Epoxy Resin to Eliminate the Risk of Sustained Casing Pressure (SCP)

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Abstract

During primary cementing, the cement coverage around casing strings from the shoe to the surface becomes critical in wells drilled across influx zones. However, the complexities of wellbore conditions sometimes prevent the cement's top from reaching the surface. In these circumstances, a "top job" is performed to fill the annulus with cement to the surface. This article outlines the successful deployment of an innovative resin system for the top job to enhance wellbore integrity.

When cement is not bonded well to casing, it can create major complications such as gas migration. The tool used to inspect bonding is called a cement bond log (CBL), which identifies areas of unbonded or "free" pipe. The issue becomes more significant if the annulus is contaminated with drilling fluids, exposing the cementing operation to the risk of sustained casing pressure (SCP). The recommended approach involves pumping through the wellhead's side-outlet valves, incorporating epoxy resin into the cement slurry. The tight injectivity conditions necessitated using a specialized portable high-pressure pump with a significantly low flow rate capability. Multiple treatments were pumped as planned to squeeze the annulus.

In tight injectivity environments, conventional cement systems often lack effectiveness for squeeze operations due to the high concentration of solids. Therefore, resin systems are the ideal choice for this type of remediation. The controlled low rheology of the resin enables deep penetration with a maximum injected volume. Carefully timed setting ensures optimal penetration and placement before the resin cures, permanently blocking potential pathways for any future influx from the wellbore. Additionally, the mechanical properties of the resin allow for improved wellbore integrity compared to traditional solutions. This solution can provide significant value by ensuring well integrity while reducing operational costs for rigs globally. The resin system not only helps operators eliminate the SCP and minimize risk but also substantially reduces their carbon footprint by eliminating the need for future remediation.

The resin system is becoming an increasingly preferred option in the industry, replacing traditional cement in numerous important remedial applications and thereby lowering the carbon footprint. This article underscores the essential laboratory testing, field implementation processes, and evaluation methods for treatments to ensure that this technology can become a vital resource for future remedial operations.

Introduction

Well cementing is used for:

- Isolate zones by preventing fluid migration between formations
- Support and bond casings
- Protect casing from corrosive environments
- Seal and hold back formation pressures
- Protect casing from drilling operations such as shock loads
- Seal loss circulation zones
- integrity experience different challenges due to:
- Opening windows for multilateral wells
- Temperature increase in production phase
- Pressure increase due to gas production
- Changing of mud weight after drilling to a lighter weight or switching to lower densities completion fluids.
- Changing of mud weight during drilling different formations

Monomers are small molecules that make up the large sized molecules named polymers. We could have different monomers bonding such as linear, branched or network structure. Monomers can have active functional groups that can be used to synthesis polymers with specific targets.[1]

Polymers can be natural or synthetic depending on their original source. We could have thermoset polymers, thermoplastic, elastomers and other types depending on their molecular forces. A good example of thermoset polymers are Resins and Bakelite. Recently, synthetic resins were used more in industry.[2]

Synthetic resins based on their molecular forces could be thermoplastic or thermosetting type. The thermosetting resins have low molecular weight and are cured by heating or chemical reaction. The conversion from liquid to solid is known as curing which is irreversible process. The curing process result in resin with improved mechanical properties since linear resins are crosslinked into three-dimensional network.[3] [4]

The curing process can be controlled by adjusting the setting or thickening time by different methods.[5]The resulted cured thermosetting resins have good mechanical properties and are resistant against corrosion and solvents. Good examples of thermo setting resins are vinyl ester resin, polyester resin, phenolic resin furan resin, and epoxy resin. Epoxy resins are superior among all the previous mentioned resins with respect of their thermal, adhesion and mechanical properties after curing.[6] [7]

They are used in different applications in different fields. They are widely used for adhesion, in oilfield applications, and for coatings and paint industry.

In Oilfield industry, epoxy resins have been widely used to support well construction and completion, and also in the area of production. Unconsolidated sands are a major issue for oil and gas wells. Epoxy resins were utilized for the strengthening of unconsolidated sands[8] to replace cement for Deepwater wells[9] and to cement casings and improve wellbore isolation for chemical disposal wells.[10] [11]

Another application in the oilfield industry are fixing casing leaks. Casing leaks are major issues for old wells in corrosive environments. We can use chemicals or mechanical methods to fix these leaks.

Casing patches and expandable tubulars are used at the leaking part of the casing to isolate it. Mechanical methods for casing repairs provide a more consistent and reliable isolation than chemical methods. However, they do have limitations. Their cost is higher than chemicals and they reduce the internal diameter of the wellbore, which makes it challenging to run downhole tools across the repaired interval.[12]

Cement is an example of chemical solution to casing leaks. We can perform squeeze operation, which is a standard operation in the oil and gas industry. Squeeze operations are used with liquid cement slurries to improve the isolation between the wellbore and the leaking casing interval. However, cement also has limitation. We cannot inject liquid cement in small leaking intervals.[13]Cements has solids that will prevent the cement from deep penetration to achieve good casing repair.

Another shortcoming of cement is its brittleness and low resistance against mechanical stresses. After using cement slurries in fixing casing leaks and after the cement become hard, the drilling operation and completion results in stressed and vibration that cause fractures within the cement, which may reduce and affect the integrity of the isolated leaked zone.

Resins have been implemented globally as an alternative fluid to cement in squeeze operations because we could use them without any solid. Due to zero solid content, the liquid resin allows deeper crack penetration and improved isolation. In general, polymer resins provide improved rheology, and mechanical properties compared to conventional squeeze cement formulations.

Resin formulations were also applied to fix casing repairs and improve wellbore integrity. Resins cross linking are affected by temperature. The temperate will expediate the speed of reaction of resin with curing agents. We always need to run intensive experimental lab work to design the required concentration of curing agents with resin at simulated downhole conditions to prevent any immature resins setting. Also, temperature affect the mechanical properties of the cured resins, as curing temperature approach glass transition temperature, the mechanical properties of the cured resin are compromised.[14]

The objectives of this paper are to compare performance of different epoxy resins and their different curing agents. The curing agents that will be investigated are aliphatic amines and aromatic amines. The physical characterization will include viscosity measurement, setting profile under simulated temperature and pressure, compressive strength development and shear bonding performance after curing at downhole conditions. In addition to that NMR, Infrared spectroscopy, DSC, SEM, CT Scan and Fluorescence spectroscopy are targeted to use to fully understand the structures, molecular weights, amorphous and crystalline phases. Once full characterization is conducted, we will be able to find best formulations of resins and resins/cement and their curing agents that are suitable for the different field applications.

Resin

We will compare performance of different epoxy resins and their different curing agents in this study. We will start first with epoxy resins that are imported from the US (Halliburton Well-Lock) and their corresponding aromatic curing agent.

The resins are Diglycidylether of bisphenol-F (DGEBF). It is a linear epoxy resin and it can be synthesized by reacting bisphenol-F with excess amount of epichlorohydrin in present of an alkali hydroxide, Fig.1.[15]

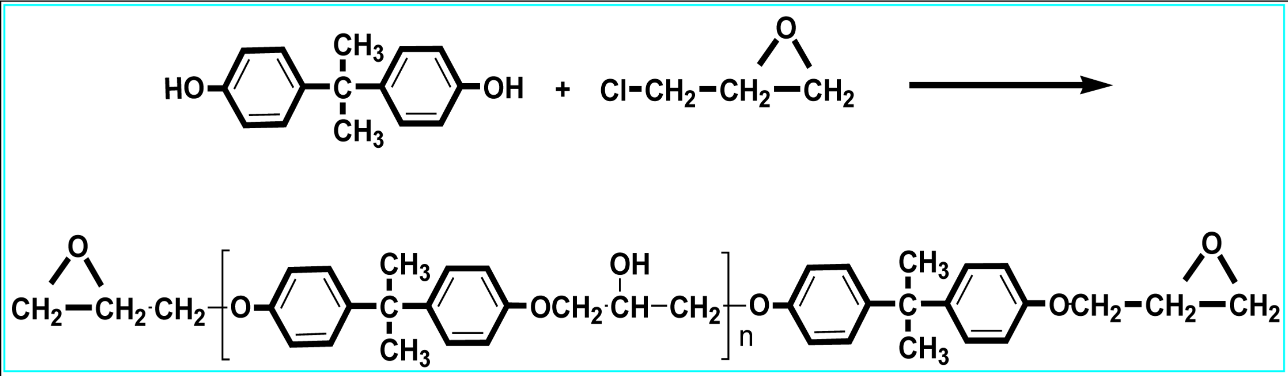


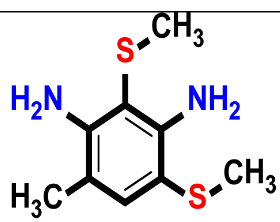
Figure 1—Diglycidylether of bisphenol-F epoxy resin[16]

This epoxy resin is then mixed with an amine curing agent convert from liquid to solid. This curing process is a step-growth polymerization. At the beginning, a longer leaner molecule will build up resulting in slow increase in viscosity. After that we will have a cross linkage of the chains with each other that result in a rapid increase in viscosity. This quick cross linking can be manipulated by the concentration and type of the curing agent as well as the chemical reaction conditions.[17] [18]

Curing Agent

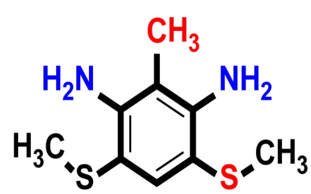
Comparing aliphatic amines to aromatic amines; shows that they are stronger and less sterically hindered which indicate the fast curing process by aliphatic amines.[19] In other words, setting time of the final epoxy resin is shorter using aliphatic amines compared to aromatic amines as a curing agent at the same conditions. However, curing epoxy resin with aromatic amines produces stronger cured resins with outstanding resistance against heat and chemical attack compared to the aliphatic ones.[20] For example, cured resins with aliphatic amines showed resistance against temperature of 100 °C; 200 °F how?. In comparison, cured resins with aromatic amines showed resistance against temperature of higher than 120 °C (240 °F). Also, less toxicity and heat generation are obtained with aromatic amines.[21]

The following aromatic amines Fig.2, is used. It is also imported product and not locally available.



a

4-methyl-2,6-bis(methylthio)benzene-1,3-diamine



b

2-methyl-4,6-bis(methylthio)benzene-1,3-diamine

Experimental

The experimental work was conducted to evaluate the rheological and mechanical properties of resin systems. The tests also were performed to evaluate the thickening time, rheological and the mechanical properties for the epoxy resins systems.[22]

The test equipment depends on the specific tests needed for the job. Some of the equipment used are ultrasonic cement analyzer (UCA), mixer, consistometer, and heating oven. The mixing procedures are conducted as follow:

1. Add the appropriate amounts of base resin (Resin 1 or Resin 2) and mix it under low shear (less than 2,000 rev/min) until the mixture is clear and homogeneous.
2. Add a curing agent to the blender and mix it under low shear condition.
3. Immediately clean all mixing equipment using solvent or acetone.
4. Turn the blender on and set it at 4,000 rev/min. Add the resin mix to the blender while it is mixing at 4,000 rev/min. After all of the resin has been added, continue mixing for 60 seconds.
5. The sample can be used for thickening time, rheology, UCA, cured samples for mechanical properties, or other required testing

Figure 3—The cured epoxy resin system[23]

Rheological properties[24]

While cement displays rheological properties resembling Bingham Plastic or Herschel Bulkley, polymer epoxy resins exhibit a Newtonian behavior for a low molecular weight. However, as the weight of molecules increases, the polymer epoxy resins systems behaves as power law fluid. In fact, the resins system builds viscosity and lacks gelling development. The resins systems enable hydrostatic pressure while in liquid. However, as the sample transfers from liquid to solid, the pressure gradient builds up until it becomes completely solid where there will be no hydrostatic pressure. The resin was able to seal gas as transferring from liquid to solid.

Rheological properties of the polymer resin systems were measured using viscometer under atmospheric conditions. Table 1 shows the rheological properties of the epoxy resin system at a temperature of 80 F. It should be noted that it is possible to prepare different resin systems with different rheological properties depending on downhole conditions.

Table 1—Rheology of epoxy resins at 80 °F

Temperature (°F)	300	200	100	6	3	PV / YP
80	205	145	73	5	3	206/ 4

Mechanical Properties Testing[25]

It is possible to modify the mechanical properties of the resin depending on downhole conditions. For the epoxy resin system, the tensile strength ranges from 100 to 2000 psi and the Poisson's ratio is close to the rubber.

Using UCA, the compressive strength of the epoxy resin system was measured as shown in Fig.4. The compressive strength of the epoxy resin ranges from 1000 to 15000 psi when tested under lab conditions.

One of the tests was done after heating the epoxy resin up to 200 F for the first 24 min and then the further heated to 225 °F. The resin was left to cure for four hours at 225 °F. It was observed that the crush compressive strength was equal to 5000 psi as shown in Table 2.

Figure 4—Ultrasonic cement analyzer

Table 2—Compressive strength test results

Time (min)	Curing temperature (F)	strength
200	250	5000

Thickening Time Test[26]

Thickening time test was conducted using API standard[27] high pressure and high temperature (HP/HT) consistometer as shown in Fig. 5. This test measures the cohesiveness and the pumpability of the polymer epoxy resin. The test results at a temperature of 200 F are shown in Table 3.

Figure 5—HP/HT consistometer

Table 3—Thickening time test results

Temperature (F)	Pressure (psi)	Batch Mix (min)	Reached in (min)	70 Bc (hh:mm)	100 Bc (hh:mm)
200	9000	45	24	4:37	4:40

Our target is to develop a low viscosity epoxy resin system capable of remediation of CCA, applicable to the temperature range of 70°F to 250°F, with a controllable curing time to form a robust but flexible sealant for long term CCA protection. Below is the main target for the developed resin formulation:

- A formulation suitable for application during remedial services for the repair of casing-casing annuli (CCA) pressure problems within wellbores.
- Develop operational guidelines for use at various temperatures and pressures.
- Product Specifications should include:
  - Temperature operational window from 70°F to 250°F
  - Low viscosity during pumping stage
  - Controlled set times
  - 1-2 hour set time at desired temperature
  - Resin should exhibit elastomeric and/or flexible properties.

Field Application

While primary cementing 9-5/8″ casing in 2-stages in Sep-2023, the DV tool's packer didn't fully inflate and couldn't hold the complete cement column in 2nd stage. A top job was performed by pumping cement from one of the side-outlets, but all pumped volume of CMT came out from the other side-outlet. So theoretically, no cement could be pushed down inside the annulus during the top job. Later, the logs confirmed free pipe from surface to 2830 ft depth. An attempt for injectivity test @ 500 psi was made in Sep-2023, however there was no injection at all. Few weeks earlier, the well's side-outlet valves for the 13-3/8″ × 9-5/8″ CCA were opened, and during opening some vacuum/suction was observed. This gave some positive hint/indication about having some cavity inside the annulus. The 13-3/8″ × 9-5/8″ annulus was filled with 98pcf OBM from 2830 ft depth all the way to surface. The plan was to inject a reliable sealing material through the Wellhead's side-outlets, utilizing Haskel pump.

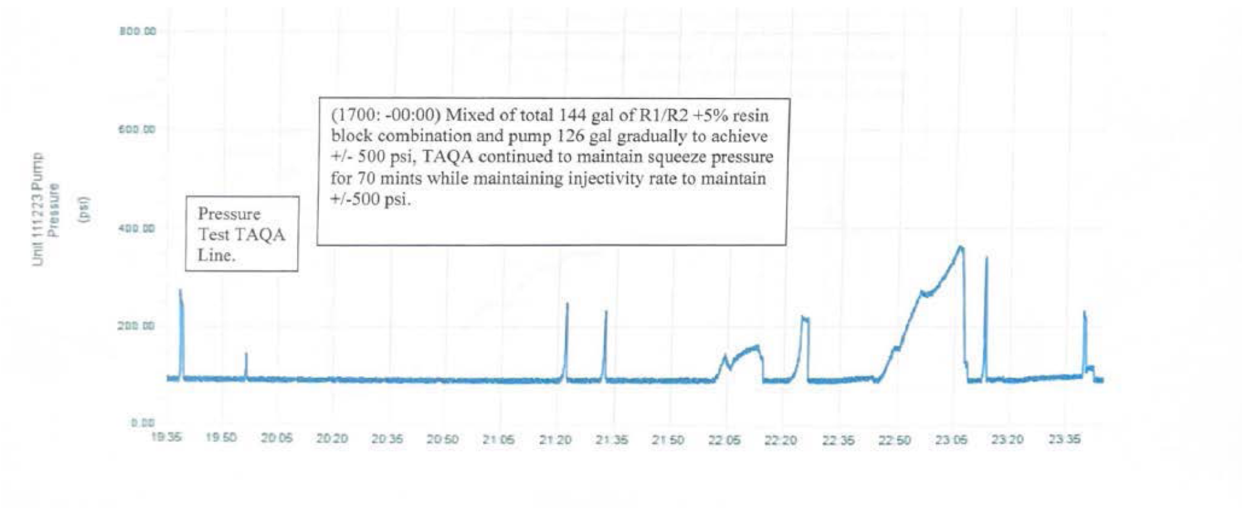
The objective was to pump a reliable sealing material inside the annulus to block the passage for any future possible influx from the wellbore. It was recommended pumping a Solid free fluid to improve the chances of success. Also it was recommended not to perform any dedicated injectivity test (i.e., using a fluid other than the cross-linked resin), due to the high possibility of very tight injectivity.

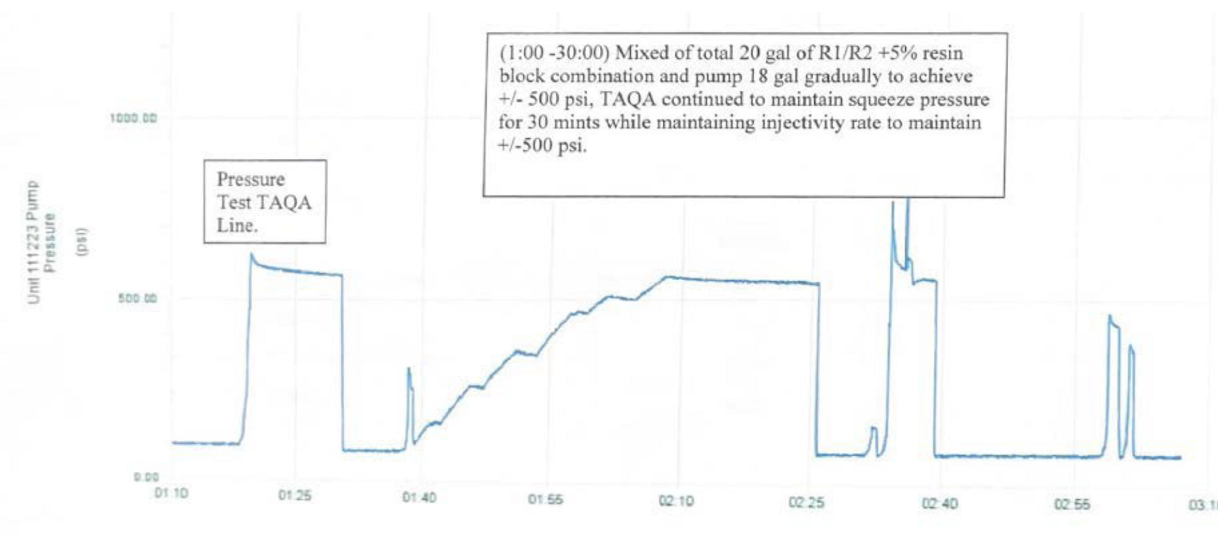
Confirmation Testing was performed onsite to decide the final formulation. Resin samples were mixed onsite using various concentrations of cross-linker to check the Gel time and final curing time. The below recipe was finalized to execute the job:

	Formula, gm	One bbl Formulation lb/bbl (42 gallons)	One bbl Formulation	
			lb/bbl (4 gallons)	kg/bbl (4 gallons)
AR-1	80	280.57	26.72	12.12
AR-2	20	70.14	6.68	3.03
Crosslinker	5	17.54	1.67	0.76

Below is the operational summary. The maximum squeeze value of 500 psi was decided.

a. The 1st treatment was performed inside 9-5/8″ × 13-3/8″ annulus. Annulus got filled-up after pumping around 3 bbl of ARC Resin. Both side-outlets were flushed with 0.25 gal (each) of non-cured resin. The wing valves were closed, and decision was made to let the Resin cured for 24 hrs. and try one more time to inject resin.
b. 2nd treatment was performed and around 4-gal Resin was pumped to fill up the annulus. After then, one of the side outlets was closed and 14-gal additional Resin was pumped while squeezing at 550 psi. Pressure remained steady for 10mins. Both side-outlets were flushed/injected with 0.25 gal (one by one) of non-cured resin and both wing valves were closed while keeping the final pressure trapped inside the annulus. Pressure was released after 24 hrs. Slight volume of resin with received in returns.





Pressure tested the annulus at 300 psi using fresh water. Around 0.25 gal of water was pumped. The pressure value remained stable, and then finally the pressure was released. Around 0.25 gal of freshwater returns were recovered. The job was deemed as successful as the maximum volume of resin was injected and squeezed into the annulus to provide a dependable barrier against any possible flow in future.

Recommendations

In order to offset this challenge effectively in future, it is recommended to run/install the three cement baskets (as contingency) in tandem at around 500 ft from surface while RIH the casing. After the primary cement job, if the DV packer does not perform as required and the level of annular fluid keeps on falling, let it fall and also let the 2nd stage cement slurry take initial set with minimum of 500 psi compressive strength. During this process, the annulus needs to be monitored very carefully to ensure that no formation influx is observed inside the annulus. Once the cement slurry develops 500 psi, then check the annulus.

If there is visible fluid level inside the annulus, then start injecting around 30-35 bbls of resin through the annulus in order to fill the entire annulus as top job. If the fluid level inside the annulus is not visible, then start injecting around 10 bbls of 118 pcf accelerated (2-3 % Calcium Chloride) cement slurry with heavy concentration of fibers followed by around 30-35 bbls of resin through the annulus in order to fill the entire annulus as top job.

Conclusions

- While primary cementing 9-5/8″ casing in 2-stages in Sep-2023, the DV tool's packer didn't fully inflate and couldn't hold the complete cement column in 2nd stage. A top job was performed by pumping cement from one of the side-outlets, but all pumped volume of CMT came out from the other side-outlet.
- The objective was to pump a reliable sealing material inside the annulus to block the passage for any future possible influx from the wellbore. It was recommended pumping a Solid free fluid to improve the chances of success.
- It was recommended not to perform any dedicated injectivity test (i.e., using a fluid other than the cross-linked resin), due to the high possibility of very tight injectivity.
- Confirmation Testing was performed onsite to decide the final formulation. Resin samples were mixed onsite using various concentrations of cross-linker to check the Gel time and final curing time.
- The job was deemed as successful as the maximum volume of resin was injected and squeezed into the annulus to provide a dependable barrier against any possible flow in future.
- This is the first study that show the use of epoxy resin as top job to eliminate potential fluids instead of using cement only.

Abbreviations

API : American Petroleum Institute
Bc : Bearden consistency
BHCT : Bottom Hole Circulating Temperature, °F
BHST : Bottom Hole Static Temperature, °F
BP : British Petroleum
BV : Bulk Volume, inch^3
BWOC : By weight of Cement
DP : Differential Pressure, psig
ECD : Equivalent Circulating Density, lb. /gal
HT/HP : High Temperature/High Pressure
ID : Inside Diameter, inch
OD : Outside Diameter, inch
PV : Plastic Viscosity, cp
PCF : Pounds per cubic foot

RP : Recommended Practices
TD : Total Depth, ft.
TDS : Total Dissolved solids, mg/l
WOC : Waiting on Cement
YP : Yield Point, lb. /100 ft.

SI Metric Conversion Factors

in. × 2.54*	E-02	= m
(°F-32) / 1.8*	E+00	= °C
Ft × 3.048*	E-01	= m
Gal × 3.785 412	E-03	= m^3
lbm × 4.535 924	E-01	= kg
Psi × 6.894 757	E-03	= Mpa
lbm/gal × 1.198 26	E-01	= S.G
bbl. × 1.58987	E-01	= m^3

*Conversions are exact.